

ISIDE: Proactively Assist University Students at Risk of Dropout

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Abstract—In this work, we present ISIDE, the prototype of a student dropout alert system integrated within Infostud, i.e., the online student portal of the Sapienza University of Rome. Our proposed solution is based on a student dropout prediction (SDP) module built from a large dataset of academic records using advanced machine learning techniques. Offline experiments show that the best-performing SDP model can detect students prone to leave the school with an F_1 score of 0.92. To further validate our prototype online, we run a pilot study on a subset of students from our School of Information Engineering, Informatics, and Statistics. This study shows that our prototype can detect students who are most likely to drop out early, as it clearly separates them from those with higher key engagement indicators.

Index Terms—AI in education, student dropout prediction, ML deployment.

I. INTRODUCTION

When students leave school before earning a degree, it becomes harder for them to get satisfactory salaries, live healthful lives, and open doors of opportunity [1]. For these reasons, student *dropout rate* is considered one of the key quality indicators of a country's education system.

Although dropout rates have fallen among some student populations over the last decade, millions of students do not finish their programs of study every year. According to a recent analysis conducted at the end of 2021 by the Education Data Initiative [2], the overall dropout rate for undergraduate college students in the United States is 40%. More generally, in 19 OECD¹ countries, 31% of students drop out of college on average. The same rate ranges from 24% in Belgium, Denmark, France, Germany, and Japan up to more than 40% in Hungary and New Zealand. In Italy – the target country of this work – a similar trend is shown in a 2018 report by the Italian National Agency for the Evaluation of Universities and Research (ANVUR), which states that about 35% of bachelor students leave the college prematurely [3].

A straightforward consequence of a high dropout rate is a low number of graduate students. Some countries are already struggling with this issue. For example, in Italy, only 28.9% of the population aged between 25 and 34 have successfully earned a university degree.

A high student dropout rate also has a detrimental effect on the return on investment in the education system and, more generally, in the national economy. Indeed, if we consider private institutions in the US, the revenue loss due to missing tuition from students that drop out amounts to a yearly average of \$16.5 billion [2]. Besides, each OECD country spends a certain fraction of its GDP on higher education (both public and private) [4]. Such investment could get partially wasted as students abandon school.

Therefore, to combat student dropout, many support systems exist to help students along their careers. For instance, mentoring/tutoring programs have proven effective in encouraging students to stay the course and eventually graduate. For programs like these to work effectively and efficiently, it is of paramount importance to: (i) early identify students who are at *high risk* of dropping out; and (ii) take appropriate actions, tailored for a specific individual, to reduce this risk to the minimum possible. Indeed, (i) is crucial for identifying target students (e.g., those who will be asked to join a tutoring program), whereas (ii) is pivotal in fostering each target student's engagement.

To address these issues, in this work we present ISIDE:² the *Infostud-Sapienza*³ *Integrated Dropout alErt* system. Inspired by previous similar proposals [5], we formulate the *student dropout prediction* (SDP) problem as a supervised learning task. More specifically, we collect data from Infostud – i.e., the online platform of the Sapienza University of Rome⁴ – which tracks the academic records of undergraduate and graduate students. Thus, we train several machine learning (ML) models to tackle the SDP task. We integrate the best-performing SDP method into the Infostud platform to implement a tool for proactively targeting students at risk of dropout.

Notice that our dataset does not contain such a ground truth, unlike existing approaches that leverage on data where the set of students who actually dropped out is known [6]. In fact, keeping track of this information accurately may be problematic unless considering trivial cases (e.g., when the student stops paying university fees). There exist several “blurry” situations, though, where a student is at *risk* of dropping out

All the authors have contributed equally to this work.

¹<https://en.wikipedia.org/wiki/OECD>

²This Italian name derives from the Latin “Isis” and indicates the ancient Egyptian goddess, mother of all gods.

³Hereinafter, we may simply refer to it as *Infostud*.

⁴<https://www.uniroma1.it/en>

despite their status might still look active (e.g., when the student is regularly enrolled in a course, but they have failed to pass all the exams attempted in the last, say, four or five sessions). Thus, data that consider dropout students only those who *evidently* abandon school might underestimate the number of individuals who are struggling or *thinking* of quitting. Therefore, training predictive models on such data could lead to unsatisfactory recall (i.e., high false negative errors), which instead is key for identifying potentially dropping-out students at an early stage. To address this issue, in this work we slightly relax the dropout condition to consider also students who are possibly disengaged or unsatisfied. We achieve that through a specific dropout labeling heuristic discussed in Section IV-A.

Moreover, to make the output of our SDP model transparent, we attach human-interpretable explanations to each prediction using a well-known model-agnostic explanation method, i.e., SHAP [7]. This capability would help human operators to communicate their decisions to students without blindly trusting machine-triggered alarms (e.g., not only tell a student that they are at risk of dropping out, but also *why*).

Finally, the validity of ISIDE is assessed online with a pilot study. Specifically, we use our SDP model to make predictions on a subset of students from our school of Information Engineering, Informatics, and Statistics. The results from this study demonstrate that our SDP prototype can effectively identify students considering abandoning their studies in that it can distinguish them from highly engaged peers. Such knowledge may, in turn, be used to take targeted actions in advance to discourage and prevent identified students from *really* dropping out.

The main contributions of our work are summarized below.

- (i) We collect and make publicly available a real-world, anonymized dataset of the academic careers of the Sapienza University's students,⁵ i.e., one of the largest European university.⁶
- (ii) We compare several predictive models that identify students at risk of dropout, using machine learning.
- (iii) We extract human-interpretable explanations for the predictions made by the best-performing SDP model.
- (iv) We develop the prototype of a novel application tool that uses the best-performing SDP model, and we integrate this tool into the existing Infostud-Sapienza platform.
- (v) We validate our prototype online by measuring key student engagement indicators.

II. RELATED WORK

One of the first contributions on student dropout prediction is provided by [8]. There, the authors compare several standard ML models; eventually, they find the best-performing method is Interpretable Classification Rule Mining 2 (ICRM2), which is a slight modification of ICRM [9] to specifically handle imbalanced datasets. However, ICRM2 might not scale to larger and more realistic use cases due to its high computational cost.

⁵<https://github.com/iside-dropout/iside>

⁶https://en.wikipedia.org/wiki/List_of_largest_universities_and_university_networks_by_enrollment

In [10], the SDP problem is investigated specifically in the context of Massive Open Online Courses (MOOCs). The authors observe that predictive models based on recurrent neural networks outperform standard ML methods.

Amongst the most recent work on SDP, [6] shows that most of the students leave their studies during the first year with a significant decrease in the dropout percentage in the next years. For this reason, the authors use first-year data only to train three different predictive models: linear discriminant analysis, support-vector machine, and random forest, with the last two reaching 87% accuracy.

More recently, an in-depth survey of the most relevant ML approaches to the SDP problem in online courses is proposed in [11]. This work widely discusses the issue of choosing the right model evaluation metrics. Indeed, the authors emphasize how accuracy alone can be misleading due to the data imbalance that is typically observed in the context of student dropout. To accommodate this well-known drawback, some works experiment with many standard ML methods, which in turn are validated using precision, recall, and F_1 score [12]. The vast majority of existing SDP methods, though, are validated using the Area Under the Receiver Operating Characteristic curve (AUROC), instead of or in addition to accuracy. Indeed, AUROC is considered a solid evaluation metric as long as the data imbalance is not highly severe.

The most similar contribution to our work is presented in [5], where authors not only build an SDP model but also integrate this tool in a platform capable of alerting in case it finds potential students at risk of dropping out. When an alarm is fired, this reaches the faculty welfare coordinators. They in turn are responsible for scheduling interventions in order to avoid reported students from dropping out. The best-performing SDP model, however, achieves 80% AUROC score and the scale of this study is way smaller than the one presented in this work (based on 75,000 students of the Sapienza University of Rome). A larger scale predictive analysis on student dropout is recently provided by [13]. Here, the authors find that including protected students' attributes (e.g., race or gender) does not impact prediction accuracy.

In conclusion, even though the SDP problem has been largely investigated under the lens of ML, as indicated by the set of relevant work mentioned above, we believe that our paper stands out from existing contributions in terms of *magnitude* and *quality assessment*. Indeed, except for [13], this work is the only one approaching the SDP problem at scale, leveraging data collected from a real-world, huge university. Moreover, to the best of our knowledge, no existing method attaches human-interpretable explanations to the predictions made by the SDP, thus effectively motivating why a student is at risk of dropping out. Finally, validating the quality of our SDP integrated tool *online* by measuring key engagement indicators both for dropping out and non-dropping out students is also unprecedented.

III. STUDENT DROPOUT PREDICTION

A. Background and Problem Formulation

Let $\mathcal{S}$ be the student population of interest. Suppose that there exists a *feature mapping* $\phi : \mathcal{S} \mapsto \mathcal{X}$, where $\mathcal{X} \subseteq \mathbb{R}^n$, namely a function that maps each student to an n -dimensional feature vector, i.e., $\mathbf{x} = \phi(s)$. Moreover, we assume that each student feature vector $\mathbf{x} \in \mathcal{X}$ is associated with a binary class label $y \in \mathcal{Y}$, by some *unknown* target function $f : \mathcal{X} \mapsto \mathcal{Y}$, i.e., $y = f(\mathbf{x})$. Usually, we define $\mathcal{Y} = \{0, 1\}$, with the convention that $y = f(\phi(s)) = 1$ indicates that s is at *risk* of dropping out, or $y = 0$ otherwise.⁷

Informally, the goal of supervised learning is to find the function h^* that best approximates the true mapping f . More specifically, given a hypothesis space $\mathcal{H}$, a sample dataset of m i.i.d. instances $\mathcal{D} = \{(\phi(s_i), y_i)\}_{i=1}^m = \{(\mathbf{x}_i, y_i)\}_{i=1}^m$, and a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \mapsto \mathbb{R}_{\geq 0}$, the supervised learning problem reduces to the following optimization task:

$$h^* = h_{\theta^*} = \arg \min_{h_{\theta} \in \mathcal{H}} \mathcal{L}_{\mathcal{D}}(h_{\theta}), \quad (1)$$

where $\mathcal{L}_{\mathcal{D}}(h_{\theta}) = \frac{1}{m} \sum_{i=1}^m \ell(h_{\theta}(\mathbf{x}_i), y_i)$. Several hypothesis spaces $\mathcal{H}$ can be considered and, depending on the specific supervised learning problem, different loss functions ℓ can be adopted. In particular, student dropout prediction is an example of a binary classification task, where (binary) *cross-entropy* loss is typically used [14].

Finally, the objective defined in (1) is usually solved via iterative numerical methods, i.e., gradient-based methods like *gradient descent* [15] or similar.

IV. METHODOLOGY

A. Data Preparation

To tackle the SDP task as a supervised learning problem, i.e., to solve the objective defined in (1), we first need to extract a sample dataset of student records $\mathcal{D}$. The data preparation stage can be further decomposed into the activities below.

a) Raw Data Extraction: Sapienza is amongst the largest and highest-ranked European university,⁸ counting around 120,000 enrolled students. *Infostud*⁹ is the official online platform that currently tracks the academic records of more than 100,000 Sapienza's undergraduate and graduate students overall.¹⁰

The Infostud platform can be accessed either via web browser or mobile app (both iOS and Android). In this work, however, we consider only Infostud Lab,¹¹ i.e., the mobile app, which is more reliable and properly instrumented, and counts 45,000 monthly active users (MAUs)¹² on average.

⁷Without loss of generality, in other contexts it is defined $\mathcal{Y} = \{-1, +1\}$.

⁸<https://www.timeshighereducation.com/world-university-rankings/sapienza-university-rome>

⁹<https://www.uniroma1.it/en/pagina-strutturale/students>

¹⁰Some students may not be properly tracked by Infostud due to several reasons (e.g., abandonment, graduation, usage of unsupported devices, etc.)

¹¹<https://www.sapienzaapps.it/apps>

¹²We define a MAU as a user who logs into Infostud at least once on a given month.

Specifically, we collect 11 months worth of mobile app data (from June 2021 to May 2022). We want to remark that this dataset contains the whole historical career of each student, not just the information limited to the time period of reference.

We exclude from our sample students who are enrolled in the Bachelor's and Master's courses offered by the school of Information Engineering, Informatics, and Statistics (i.e., approximately 6,000 students), as those will be part of our online pilot study (see Section VIII). Still, we end up with a collection S of about 90,000 students.

b) Data Cleaning: After raw data extraction is complete, we are able to represent each student $s \in S$ with a record of 30 attributes available from Infostud. Before we can feed our data to the next stages of the ML pipeline, we perform a number of preprocessing steps as described below.

First, we remove from our dataset S the following entries:

- students who have no information except their matricula;
- students who are about to graduate;
- incoming Erasmus students, as they will spend only six months with Sapienza and therefore it makes no sense to use them for training any dropout prediction model;
- students of the School of Medicine because they have some peculiarities (e.g., 1-credit exams, 6-year course instead of the usual 2, 3 or 5 years) that may inject undesired bias into the learned SDP model.

We further discard students whose birth or matriculation date sounds unrealistic. In particular, we retain only students born after January 1, 1970, and enrolled after January 1, 2000.

Eventually, as a result of data cleaning, our final collection of students S reduces by roughly 20%, i.e., from about 90,000 to approximately 70,000 unique students, who are labeled according to the strategy described below.

c) Data Labeling: It is worth remarking that no explicit confirmation of a student dropout event is ever recorded on Infostud, thus no binary class label is naturally associated with each student, as instead required by supervised learning tasks. Besides, assigning such label manually to tens of thousands students would be infeasible. Therefore, we need to design an automatic labeling strategy for building our reasonable ground truth of students who are *likely about* to drop out. We achieve this goal using the procedure described in Algorithm 1. The final labeled dataset is: $\mathcal{D} = \{(\phi(s), y) \mid (s, y) \in S^{\mathcal{Y}}\}$, where ϕ indicates the student's feature representation. The list of features extracted is shown and described in Table I. Obviously, the labeling strategy (i.e., the threshold δ) has an impact on the resulting ground-truth dataset, and therefore on the downstream task, i.e., student dropout prediction. Indeed, we analyze this aspect in a dedicated study, detailed in Section V-B. In this work, we set the threshold $\delta = 6$ months, which produces a moderately imbalanced dataset $\mathcal{D}$, where roughly 75% of the students are labeled as active ($y = 0$) and the remaining 25% as potentially dropping out ($y = 1$).

Finally, we randomly split $\mathcal{D}$ into two disjoint portions – $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ – using stratified random sampling, so as to preserve the same class label distribution observed in the whole dataset. The former accounts for 80% of the instances

Category	Feature	Type	Description
Profile	programYear	numerical	Program year (e.g., 1 is the first year of the program)
	age	numerical	The student's age (in years)
	citizenship	categorical	The country where the student was born
	hasPicture	boolean	Indicates if the student was identified using the picture he/she uploaded to Infostud
	firstEnrollment	numerical	The time elapsed since the first enrollment (in years)
	lastEnrollment	numerical	The time elapsed since the last enrollment (in years)
	sex	boolean	The student's gender
	studentType	numerical	Indicates the type of the student's curriculum
	income	numerical	Annual income
	HSDiplomaScore	numerical	The high school diploma score
Academic	programCode	numerical	Internal code associated with the program
	programType	numerical	Code associated with the type of the program
	totalCredits	numerical	Total number of credits required by the program
	school	categorical	School in which the student is enrolled
	enrollmentType	categorical	Indicates whether the student has completed exams within set time period or not
	creditsBooked	numerical	Number of booked credits
	creditsEarned	numerical	Number of credits currently earned
	lastBookedExam	numerical	The time elapsed since the date of the last booked exam
	lastPassedExam	numerical	The time elapsed since the date of the last passed exam
	avgGrade	numerical	Average grade according to the Italian academic grading system [16]

TABLE I: The final list of student features extracted from Infostud.

Algorithm 1: Dropout Labeling Heuristic

```

1: Input:
2:  $S$   $\triangleright$  The set of students sampled from Infostud
3:  $\delta$   $\triangleright$  The time window used to determine the label
4: Output:
5:  $S^{\mathcal{V}}$   $\triangleright$  The set of sampled students and their dropout labels

6: procedure dropoutLabeling( $S, \delta$ ):
7:    $S^{\mathcal{V}} = \emptyset$ 
8:   for  $s \in S$  do
9:     if ( $\text{today} - s.\text{lastLogin} < \delta$ ) then
10:      # The elapsed time since the last login is within  $\delta$ 
11:       $S^{\mathcal{V}} = S^{\mathcal{V}} \cup \{(s, 0)\}$ 
12:     else
13:       if ( $s.\text{creditsEarned} \geq (\text{totalCredits} - 30)$ )
14:         # The student has completed their studies
15:          $S^{\mathcal{V}} = S^{\mathcal{V}} \cup \{(s, 0)\}$ 
16:       else
17:         # The student is at risk of dropping out
18:          $S^{\mathcal{V}} = S^{\mathcal{V}} \cup \{(s, 1)\}$ 
19:       end if
20:     end if
21:   end for
22:   return  $S^{\mathcal{V}}$ 
23: end procedure

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(i.e., $\approx 55,000$ students) and is used for training ML models, whereas the latter contains the remaining 20% (i.e., $\approx 15,000$ students) and is used for testing. This 80÷20 split ratio is a common best practice to avoid the risk of overfitting.

B. Data Exploration

Without claiming to be exhaustive, we consider three distinct factors that may affect the propensity to dropout: *academic*, *socio-economical*, and *geographical*.

Concerning the first one, in Fig. 1 we visualize the distribution of the average grade obtained by the two groups of students, i.e., dropout vs. non-dropout. As one might expect, the performance of students who are likely to abandon look different from that of other peers, and generally worst (i.e.,

the median average grade is 25/30 and 26/30, respectively). To confirm this intuition, we run a two-sample Kolmogorov-Smirnov test that indicates the underlying average grade distributions of the two groups are indeed statistically significantly different at the $\alpha = 0.01$ significance level ($p\text{-value} \ll 0.01$), i.e., they come from two different student populations. Moreover, we run a one-sided two-sample t -test to check the null hypothesis of the average grade of dropout students being greater than that of non-dropout ones. Result shows that there is enough evidence to reject the null hypothesis ($p\text{-value} \ll 0.01$). We therefore analyze the possible impact of a socio-

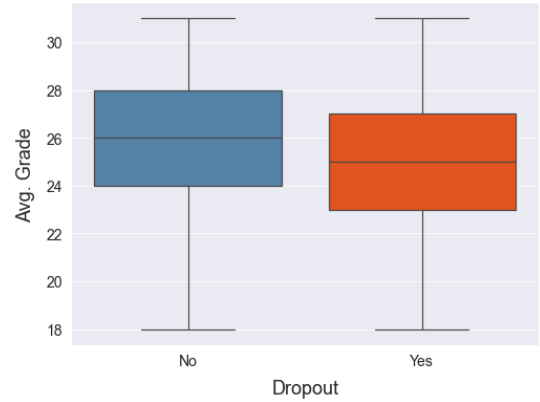


Fig. 1: The average grade distribution of dropout and non-dropout students.

economical factor. Specifically, we examine the distribution of annual income of dropout and non-dropout students. We run the same two-sample Kolmogorov-Smirnov test as above, and despite the two distributions depicted in Fig. 2 look apparently similar, there is still enough statistical evidence to reject the null hypothesis ($p\text{-value} \ll 0.01$). The same evidence is found when we run the one-sided two-sample t -test to assess if the average annual income of dropout students is greater than that of non-dropout ones. This may indicate that the annual income observed in the two groups of students is indeed different

(smaller for dropout students) and possibly a significant factor affecting the risk of abandonment. Finally, we look at the

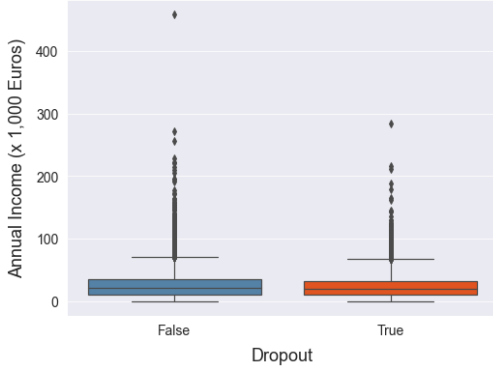


Fig. 2: The annual income distribution of dropout and non-dropout students.

geographical distribution of dropout and non-dropout students. Fig. 3 shows that all the students have a similar inclination to abandon their studies independently of their citizenship, i.e., $\approx 29 \div 30\%$ for Italian (“ITA”), European (“UE”) and non-European (“Extra-UE”) students. This is also confirmed by the result of a two-sample z-test for proportions, which highlights that the difference between the dropout ratio amongst the three groups is *not* statistically significant (p -value > 0.41).

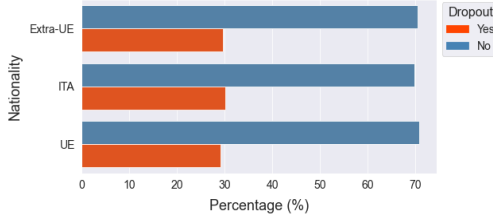


Fig. 3: The geographical distribution of dropout and non-dropout students.

C. Model Training

To implement our SDP module, we consider the following ML models: decision tree (DT), random forest (RF), gradient boosting decision tree (GBDT), logistic regression (LR), support-vector machine (SVM), and multilayer perceptron (MLP), i.e., a class of feed-forward neural network. We train each model on $\mathcal{D}_{\text{train}}$ and fine-tune its hyperparameters via successive halving grid search using 5-fold cross-validation.

RF: We test with $n_{\text{estimators}}=[250, 500, 1000]$ each estimator having $\text{max_depth}=[2, 4, 8, 12]$ and each leaf $\text{min_samples_leaf}=[16, 32, 64, 128]$.

GBDT: In addition to the hyperparameters tested for RF, we consider two loss functions ($\text{loss}=[\text{deviance}, \text{exponential}]$) and $\text{learning_rate}=[0.01, 0.05, 0.1, 1, 10]$.

LR: We compare several optimizers ($\text{solver}=[\text{lbfgs}, \text{newton-cg}, \text{liblinear}]$) running for a range of iterations ($\text{max_iter}=[50, 100, 1000]$), $C=[0.1, 0.5, 1, 50, 100]$ and class importance ($\text{class_weight}=[\text{balanced}, \text{None}]$).

SVM: We use a standard radial basis function kernel

with two possible choices of coefficient ($\text{gamma}=[\text{auto}, \text{scale}]$), several strengths of the regularization parameter ($C=[0.1, 0.5, 1, 50, 100]$), and distinct class importance ($\text{class_weight}=[\text{balanced}, \text{None}]$).

MLP: This is a 2-layer fully-connected neural network with a sigmoid activation function in the last (output) layer. We tune the MLP architecture by experimenting with several numbers of neurons in each hidden layer ($\text{units}=[16, 48, \dots, 512]$). We also apply regularization through random dropout using 10 distinct probability values ($\text{drop}=[0.0, 0.1, \dots, 0.9]$). The resulting neural network is trained by minimizing binary cross-entropy loss using Adam optimizer [17] and multiple values of learning rate ($\text{learning_rate}=[0.0001, 0.001, 0.01]$).

We evaluate the performance of each combination of model/hyperparameters above by measuring the area under the ROC curve (AUROC) averaged across the 5 folds. The best combination of hyperparameters for each model along with the obtained cross-validation AUROC is shown in Table II.

Model	Best Hyperparameters	CV AUROC
RF	$\{\text{n_estimators}=1000, \text{max_depth}=8, \text{min_samples_leaf}=16\}$	0.98
GBDT	$\{\text{n_estimators}=500, \text{max_depth}=8, \text{min_samples_leaf}=16, \text{loss}=\text{exponential}, \text{learning_rate}=0.01\}$	0.98
LR	$\{\text{solver}=\text{liblinear}, \text{max_iter}=50, C=1, \text{class_weight}=\text{balanced}\}$	0.93
SVM	$\{\text{gamma}=\text{auto}, C=50, \text{class_weight}=\text{balanced}\}$	0.96
MLP	$\{\text{units}=368, \text{drop}=0.1, \text{learning_rate}=0.001\}$	0.94

TABLE II: Cross-Validation AUROC obtained with the best hyperparameter setting of each model.

In the end, we re-train all the models with the best hyperparameters on the whole training set $\mathcal{D}_{\text{train}}$, and we measure their performance on the test set $\mathcal{D}_{\text{test}}$ previously held out. The results of this assessment are presented below.

V. OFFLINE EVALUATION

A. Model Evaluation

In Fig. 4, we compare the quality of all the best-performing models trained as discussed above, by measuring their (macro-average) *precision*, *recall*, and F_1 score on the held out test set $\mathcal{D}_{\text{test}}$, using 0.5 as classification threshold. The plot indicates that tree-based ensemble methods (RF and GBDT) outperform other approaches, with GBDT exhibiting the highest F_1 score (0.92). LR and SVM, instead, perform worse. Evidently, they are both worst in capturing the non-linearity between the input feature space and the output response. In this regard, MLP behaves similarly as SVM, although it does not match the performance of tree-based methods (F_1 score = 0.88). We believe there might be room for improvement on MLP’s performance by experimenting with its architecture design (e.g., adding more hidden units). However, this goes beyond

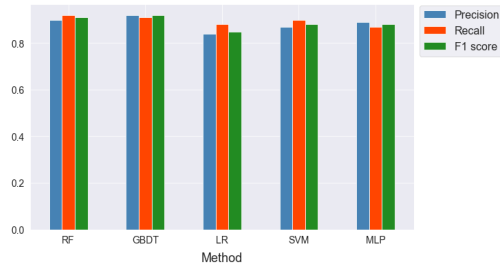


Fig. 4: Overview of all the best-performing models.

the main scope of this work, and we plan to investigate it further in future work. Besides, it is also worth remarking that tree-based models are generally easier to interpret and explain than complex neural networks [18]–[20]. Thus, when the predictive accuracy is similar, tree-based methods are generally preferable over more complex neural networks, especially if non-experts use the model’s output to make decisions, like in our scenario. For example, GBDT (as well as RF) is able to rank features according to their “predictive importance”. Fig. 5 shows the top-10 most important features according to our GBDT model. We can see that the score of the last enrollment

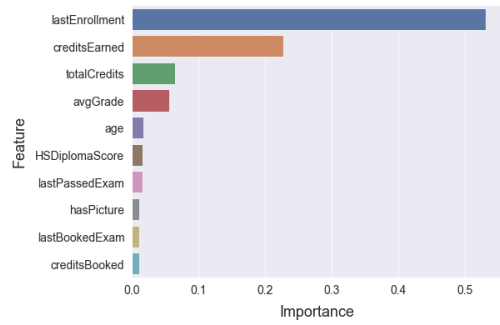


Fig. 5: Ranking of feature importance according to GBDT.

has extreme predictive power. This was not so expected since, from our preliminary data exploration, the last enrollment date alone was not highly correlated with the class label. Also, the credits earned by the students and their average grade is key to determining their level of engagement.

In the following, we select GBDT as our reference best-performing SDP model.

B. The Impact of Labeling Strategy

The automatic labeling strategy adopted to build our training set may affect the performance of any of the SDP models above. In this section, we analyze the impact of Algorithm 1 on the best-performing SDP model selected. In particular, we experiment with different values of the parameter δ , which denotes the time window used by the algorithm to distinguish between engaged students and those at risk of dropping out. In addition to the original value of 6 months used, we vary δ by setting it at 2, 4, and 9 months. Thus, we train our best SDP model on the $\mathcal{D}_{\text{train}}$ obtained with the corresponding labeling.

It is worth noticing that it would not have made sense to test values of δ larger than 11 months because our dataset traces

back the last login information no further than 11 months (i.e., from the end of June 2021 to May 2022). Thus, any values of δ greater than 9 months would trivially lead us to label almost all students as non-dropout. On the other hand, values of δ smaller than two months would have resulted in many students being labeled as dropout.

In general, we observe that the predictive accuracy of our SDP model is robust with respect to the choice of the threshold δ . More specifically, the AUROC is stable at 0.98, even when we adopt the more aggressive labeling strategy with a slight increase with stricter strategies.

VI. EXPLAINING DROPOUT RISK PREDICTIONS

The predictive accuracy of our best-performing SDP model is undoubtedly remarkable. However, this alone may not be enough for an ML-based student dropout alert system deployed in a real-world setting to be successful. In addition to its predictive power, our SDP model should also be *explainable*, i.e., it should provide human-interpretable motivations behind its output predictions. This capability would help human operators to communicate their decisions to students without blindly trusting machine-triggered alarms (e.g., not only can operators tell a student that they are at risk of dropping out, but also *why*). Indeed, the increasing demand for more explainable ML models has become prominent in many social domains like finance [21] and healthcare [22]. We believe that education is another critical domain where ML-based decision support services like our proposed SDP tool cannot ignore human-to-human relationships.

A large body of work on *eXplainable AI* (XAI) has flourished in recent years [23]. The most popular approaches can be categorized into *model-specific* [18], [20], [24], [25] and *model-agnostic* [7], [26]–[29]. Here, we consider a well-known model-agnostic explanation method (SHAP [7]).

SHAP (*SHapley Additive exPlanations*) is a game theoretic approach to explain the output of any ML model. It connects optimal credit allocation with local explanations using the classic Shapley values from game theory and their related extensions. Fig. 6 shows a density scatter plot of SHAP values

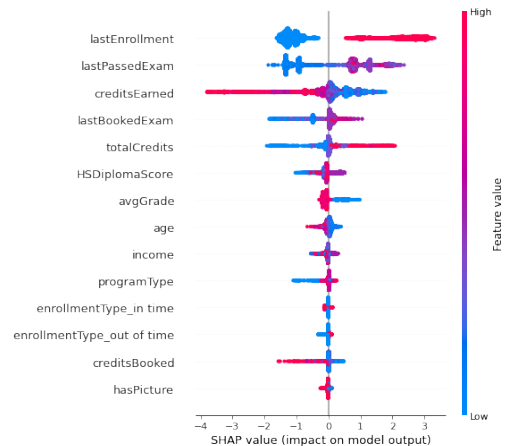


Fig. 6: Impact of each feature on the predicted risk of dropout measured with SHAP value.

for each feature to identify how much impact each predictor has on the model output. Features are sorted by the sum of the SHAP value magnitudes across all samples.

From this plot, we can make the following observations. The longer time that has passed since the last enrollment date, the higher the chance for a student to drop out. This is quite intuitive: a student whose last enrollment date is far away is very likely disengaged and therefore prone to drop out. Similarly, the fewer credits a student earns or books, the higher their propensity to abandonment. Another explanation of why our SDP model labels a student at risk of dropout is the lower average grade obtained. Overall, these insights will make any decision taken by human operators based on the SDP's output more fair and transparent to the student.

Finally, it is worth noticing that the aim of this analysis is fundamentally different from the feature ranking reported above in Fig. 5. Roughly speaking, impurity-based importance used for feature ranking summarizes the overall contribution of a feature for splitting tree nodes; SHAP values, instead, explain how much would changing a feature value affect the prediction (not necessarily correct).

VII. IMPLEMENTING OUR PROTOTYPE: ISIDE

The Infostud mobile app relies on a gateway system that exposes an API (Application Program Interface) to access data regarding students. We exploit this gateway to gather data to clone career information and obtain a data feed that we use to run our prototype. The data feeder extracts data once a week. ISIDE works asynchronously in batches: the predictive model runs on the full dataset that we gather weekly, and returns the list of student IDs with their dropout probability. We use this list in our pilot study (see Section VIII).

Due to the sensitive nature of the collected students' information, we have to tackle different problems concerning data confidentiality. On the one hand, we want to fully exploit our dataset across every stage of the ML pipeline to build the best-performing SDP model. On the other hand, we have to preserve students' privacy. Thus, we design our data feeder using a privacy-first approach, minimizing the amount of data we extract. The feeder removes all personal data in the early stages of the data extraction pipeline, replacing them with generic values (e.g., the year of birth instead of the entire date of birth) or empty values when the information is not needed. Also, student IDs are replaced with random values consistently across the dataset.

To design the prediction pipeline for the pilot study and further integration, we extend our data feeder to provide a way to de-anonymize the prediction result. We achieve that by storing the mapping between the real student ID and the random number to retrieve the original student ID. However, a confidentiality issue may arise if a malicious actor can access the mapping and the anonymized data. To prevent it, we only store the mapping encrypted with a key that we keep on a physically different system. Also, the key is accessible only by the data feeder and the server that runs the prediction model. In this way, we can preserve the confidentiality of the dataset.

Finally, as with any ML model deployed in a production environment, another key challenge for ISIDE is to effectively counter the model aging (i.e., how frequently we should re-train our SDP model on fresh data). We expect to address this problem by re-training the SDP model at least once a year to promptly consider all possible changes that may occur to student cohorts. Furthermore, although the SDP model can be continuously queried once trained and deployed, we plan to generate a full report at the end of each semester with all the highest dropout risk students. This way, ISIDE can leverage students' most up-to-date progress to make predictions.

VIII. ONLINE PILOT STUDY

To validate our prototype of the Infostud dropout alert system online, we designed a pilot study restricted to a subset of students of our School of Information Engineering, Informatics, and Statistics. Our SDP model was *not* trained on data from those sampled students, who are genuinely unseen input examples.

We let our SDP model rank the students above according to the predicted probability of dropping out. Hence, we consider two different groups of students in this ranked list: the first one contains the 50 students with the highest dropout probability (*top-50*); the second group, instead, is composed of the 50 students with the lowest probability of dropout (*bottom-50*).

Then, based on the insights discussed in Section VI, we define the following three measures, which represent key student engagement indicators, for validating the result of our online pilot study: elapsed time (in months) since the last access to Infostud (LA); elapsed time (in months) since the last exam passed (LP); and average exam grade (AG).

Intuitively, *top-50* and *bottom-50* represent students from two distinct underlying populations, i.e., high and low risk of abandonment, respectively. Therefore, we expect LA and LP to be statistically significantly larger as measured for *top-50*, whereas AG should be higher for *bottom-50*.

To validate our claims above, we run the one-sided two sample *t*-tests reported in Table III.

Results show that the sample mean of LA and LP computed on the two groups are statistically significantly different at a confidence level $\alpha = 0.01$. Specifically, while the mean LA (LP) is around 12 months (16 months) for *top-50* high dropout risk students, this falls to about 2.5 months (4 months) for *bottom-50*. On the other hand, we cannot draw a similar conclusion for AG; apparently, the mean AG computed from the two groups of students does not unveil any statistically significant difference.

These findings confirm that ISIDE can effectively distinguish between highly and poorly engaged students.

IX. CONCLUSION AND FUTURE WORK

In this work, we presented ISIDE, the prototype of a student dropout alert system integrated within Infostud, i.e., the online student portal of the Sapienza University of Rome. Our proposed solution is based on a student dropout prediction (SDP) module built from a large dataset of academic records using

Last Access (LA)			Last Passed (LP)			Average Grade (AG)		
μ_{LA}^{top}	μ_{LA}^{bottom}	$H_0 : \mu_{LA}^{top} \leq \mu_{LA}^{bottom}$	μ_{LP}^{top}	μ_{LP}^{bottom}	$H_0 : \mu_{LP}^{top} \leq \mu_{LP}^{bottom}$	μ_{AG}^{top}	μ_{AG}^{bottom}	$H_0 : \mu_{AG}^{top} \geq \mu_{AG}^{bottom}$
11.96	2.46	$p\text{-value}=2.72e-57(*)$	16.17	4.14	$p\text{-value}=7.89e-17(*)$	22.94	22.50	$p\text{-value}=0.98$

TABLE III: Statistical significance tests for comparing the differences between the three key student engagement indicators: LA, LP, and AG. For each indicator, we compute its mean (μ) in the two groups: high-risk (*top*) vs. low-risk (*bottom*) of dropout. Then, we run a one-sided two sample t -test to assess if there is enough evidence to reject the null hypothesis (H_0).

advanced machine learning techniques. Offline experiments showed that the best-performing SDP model could detect students prone to leave the school with an F_1 score of 0.92. In addition, we extracted human-interpretable explanations for the predictions made by the SDP model using state-of-the-art techniques to make any decisions taken more transparent to the affected students. To further validate ISIDE online, we ran a pilot study on a subset of students, demonstrating that our prototype could detect students who are most likely to drop out early, as it separates them from those with higher key engagement indicators.

This work paves the way for several research directions that should be further explored in future work. Amongst them, it is worth mentioning: (i) experimenting with more complex SDP modeling techniques (e.g., deeper neural networks); (ii) exploiting additional signals from the Infostud platform (e.g., social connections between students) to improve SDP accuracy; (iii) scaling up the online pilot study to other Departments and Schools; and (iv) running a large-scale student survey to validate more accurately our prototype online.

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